

Indian Diseases Dataset Analytics Report

Dataset: indian_diseases_dataset_analysis.csv

Total Records After Cleaning: 20000

Exact Duplicates Removed: 0

Columns Dropped (>4 Nulls): 4

Numeric Features Used: 6

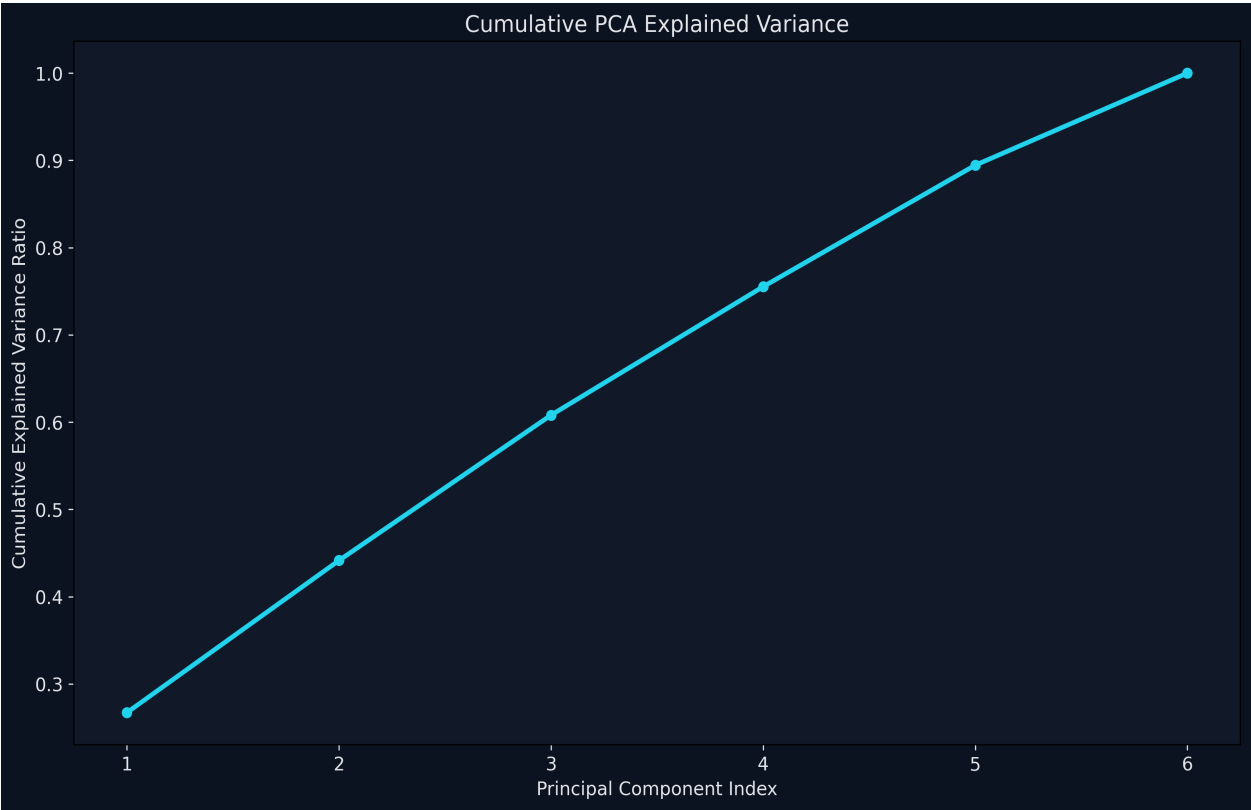
Detected Anomalies: 1000

Silhouette Score: 0.218

Total PCA Explained Variance: 100.0%

Execution Time: 55.28 seconds

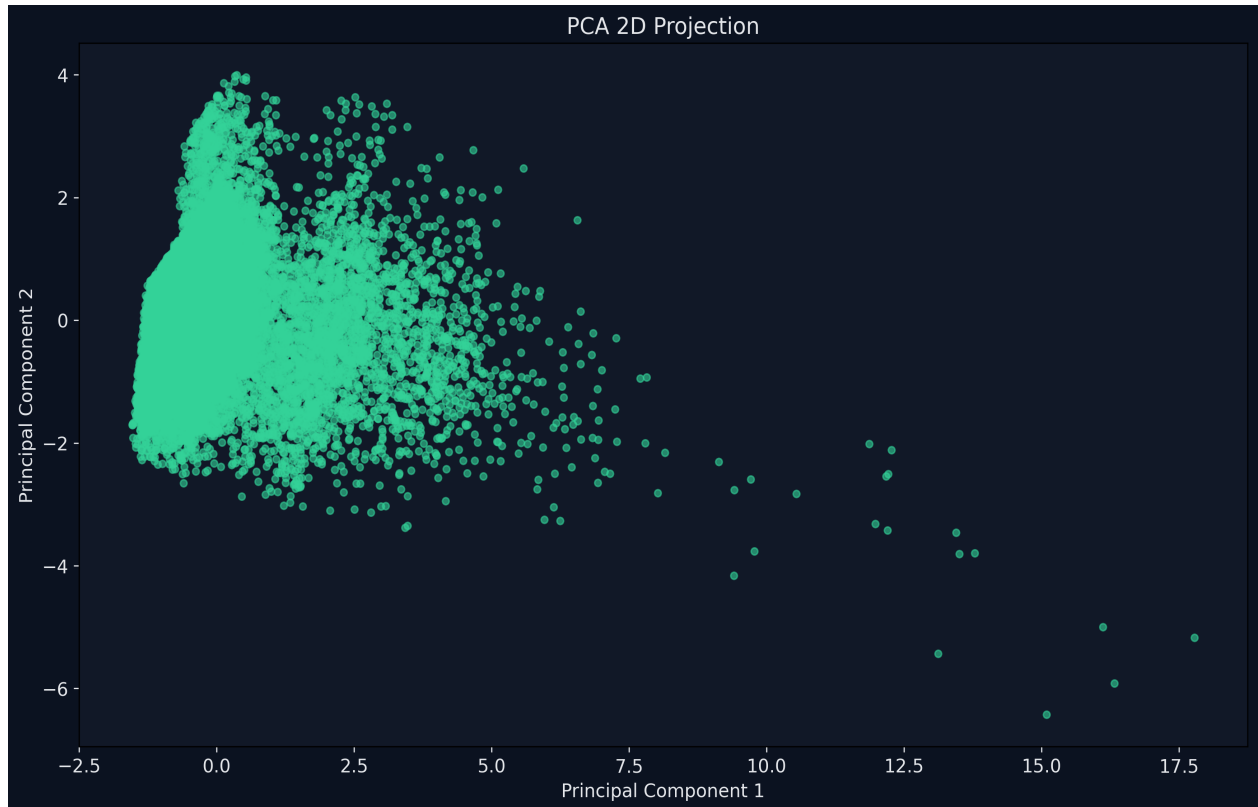
Cumulative PCA Explained Variance



Axes: X-axis: Principal Component Index | Y-axis: Cumulative Explained Variance Ratio

Explanation: Variance accumulation across principal components showing dimensional compression efficiency.

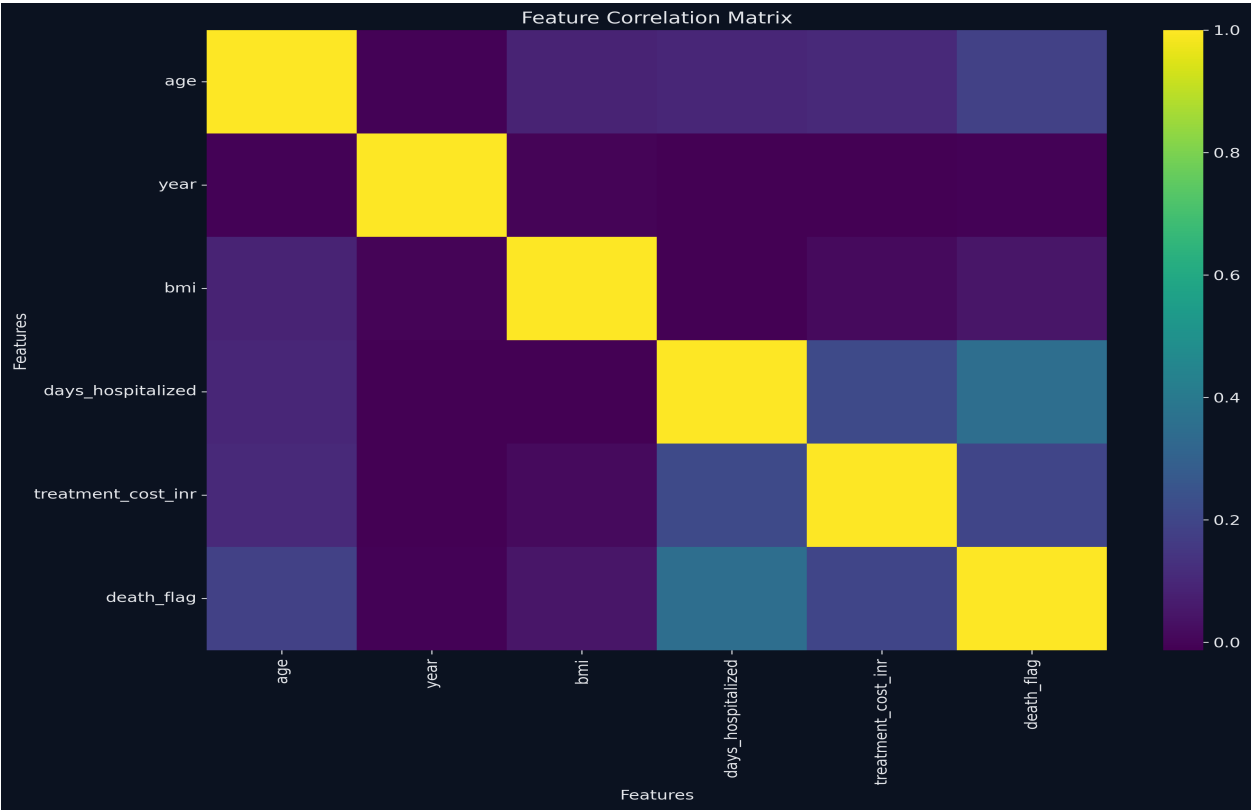
PCA 2D Projection



Axes: X-axis: Principal Component 1 | Y-axis: Principal Component 2

Explanation: Reduced dimensional representation highlighting structural variation and clustering tendency.

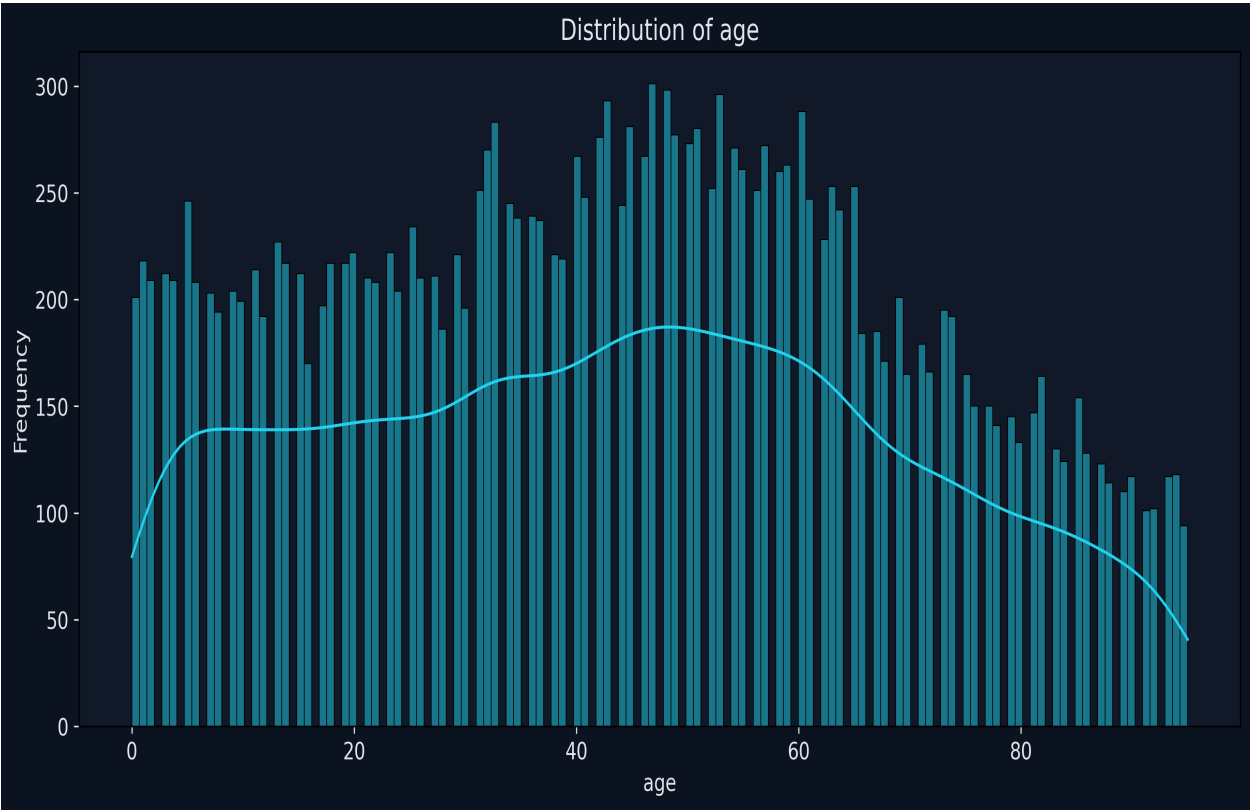
Correlation Matrix



Axes: X-axis: Features | Y-axis: Features

Explanation: Pairwise linear relationships between numeric variables to identify dependency structures.

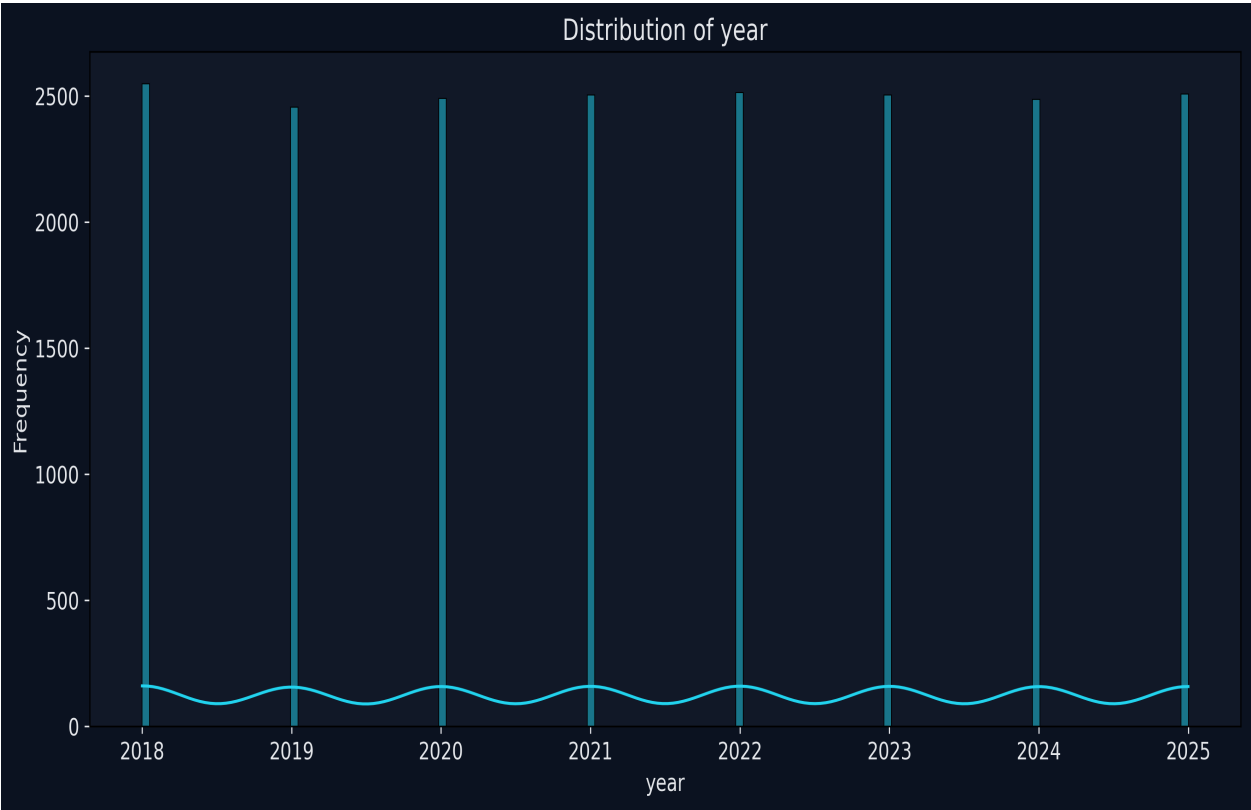
Distribution of age



Axes: X-axis: age | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

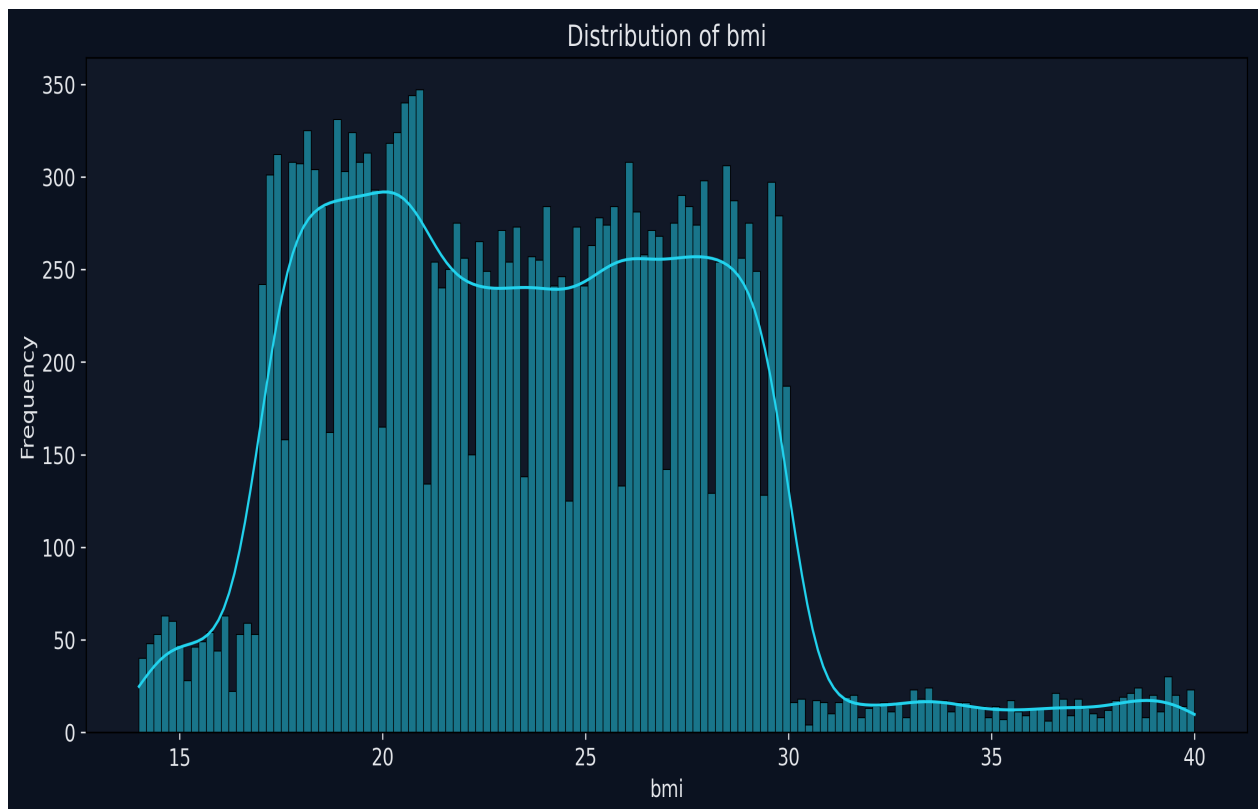
Distribution of year



Axes: X-axis: year | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

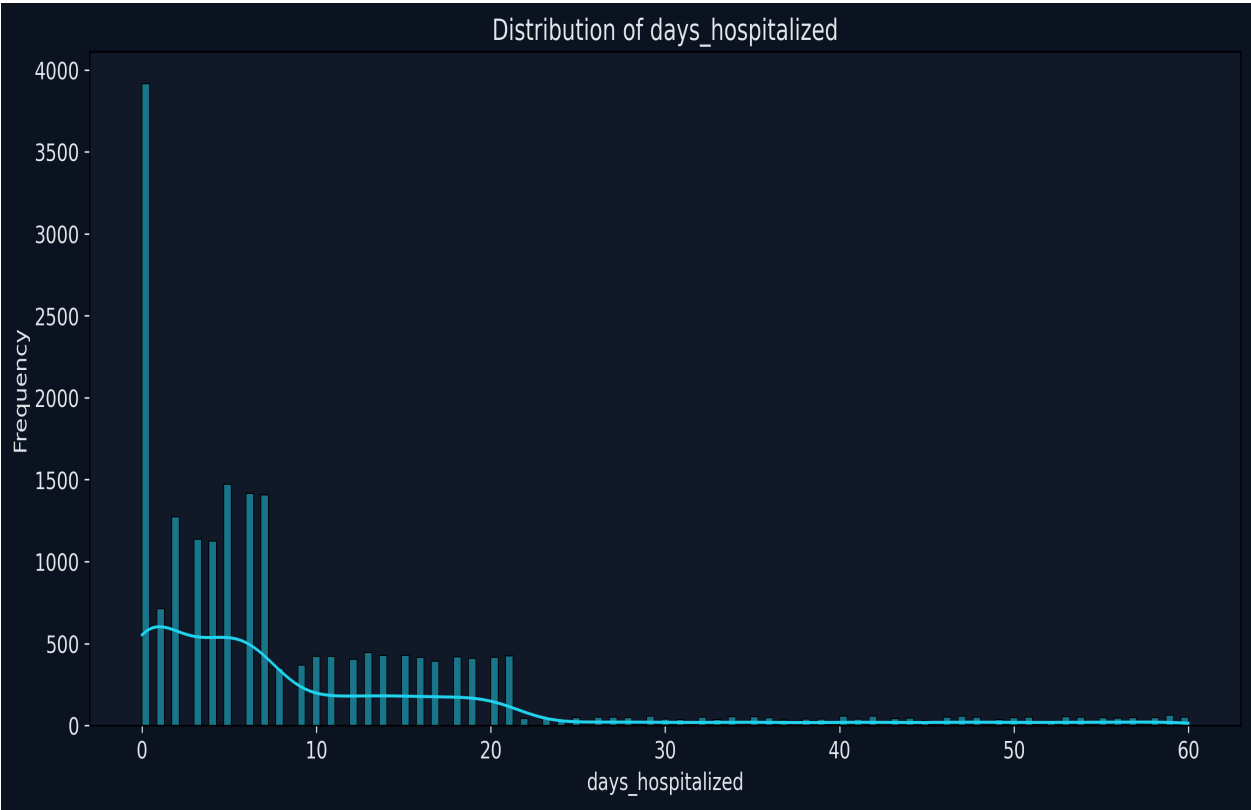
Distribution of bmi



Axes: X-axis: bmi | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

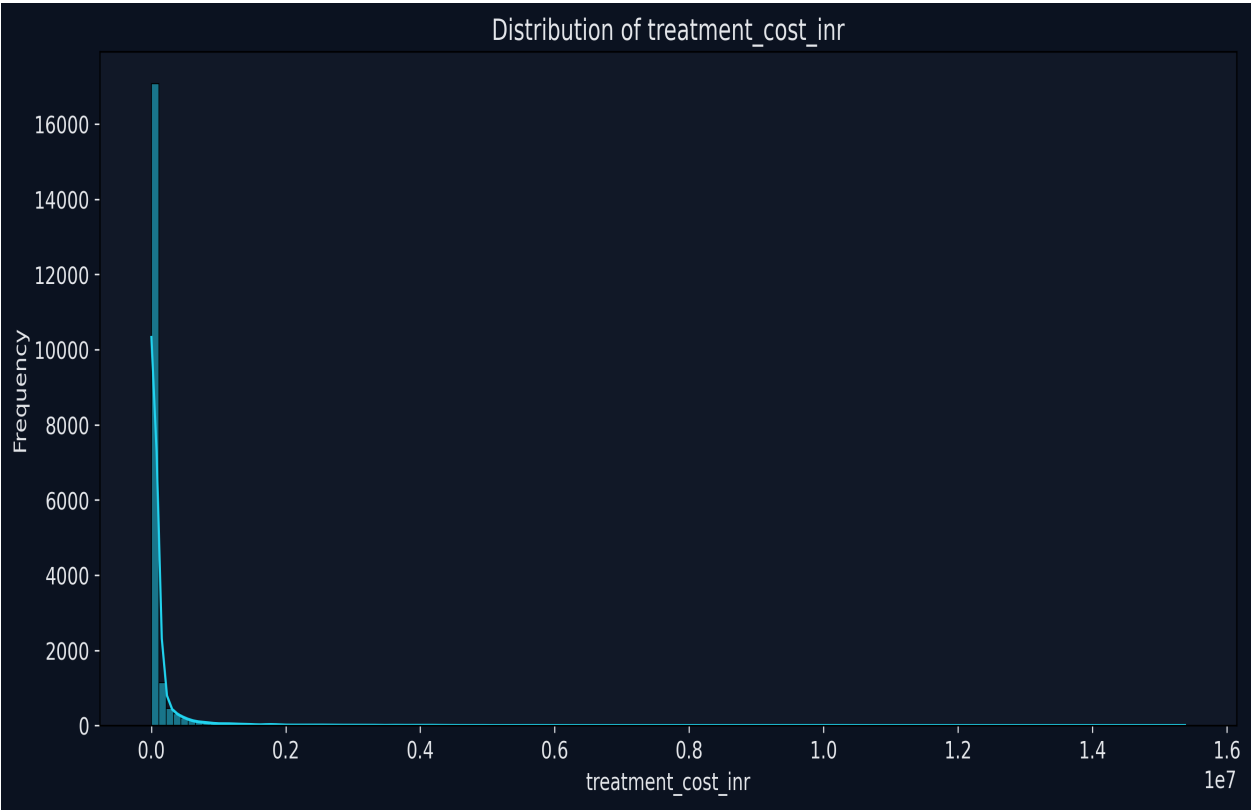
Distribution of days_hospitalized



Axes: X-axis: days_hospitalized | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

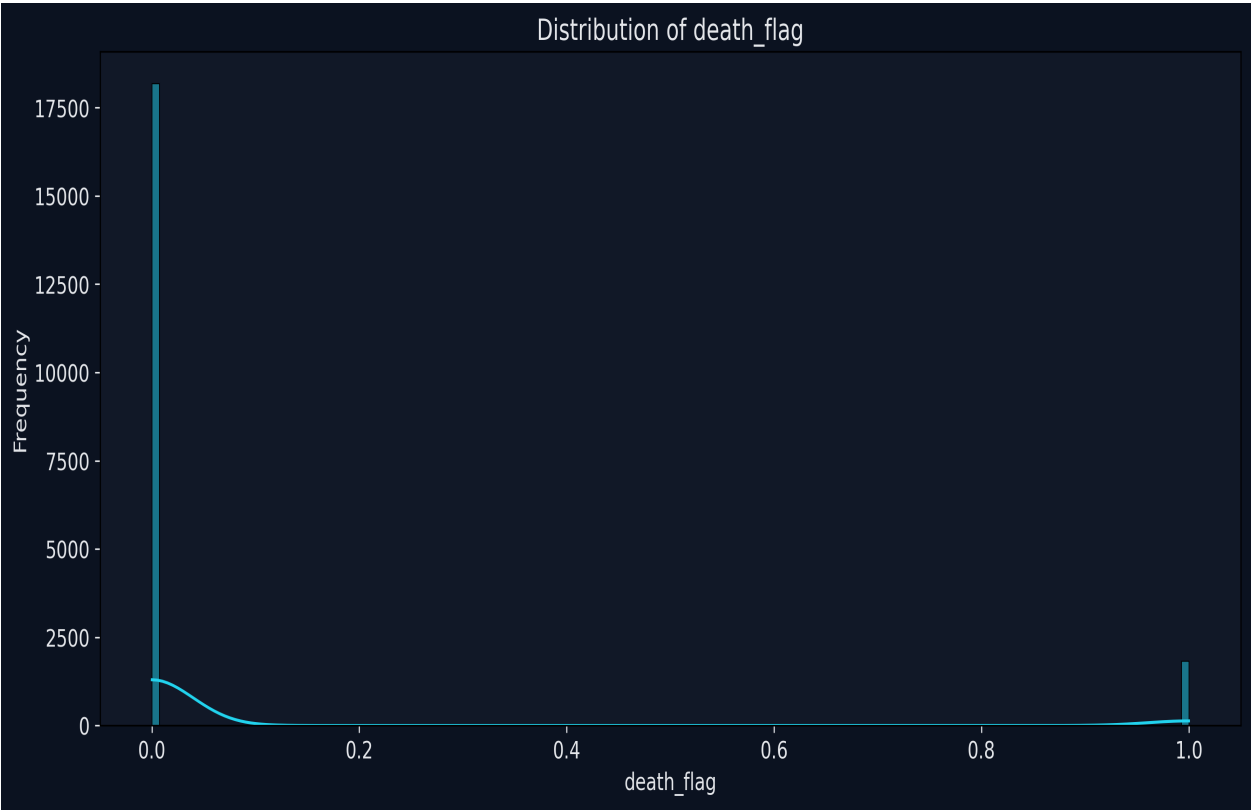
Distribution of treatment_cost_inr



Axes: X-axis: treatment_cost_inr | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

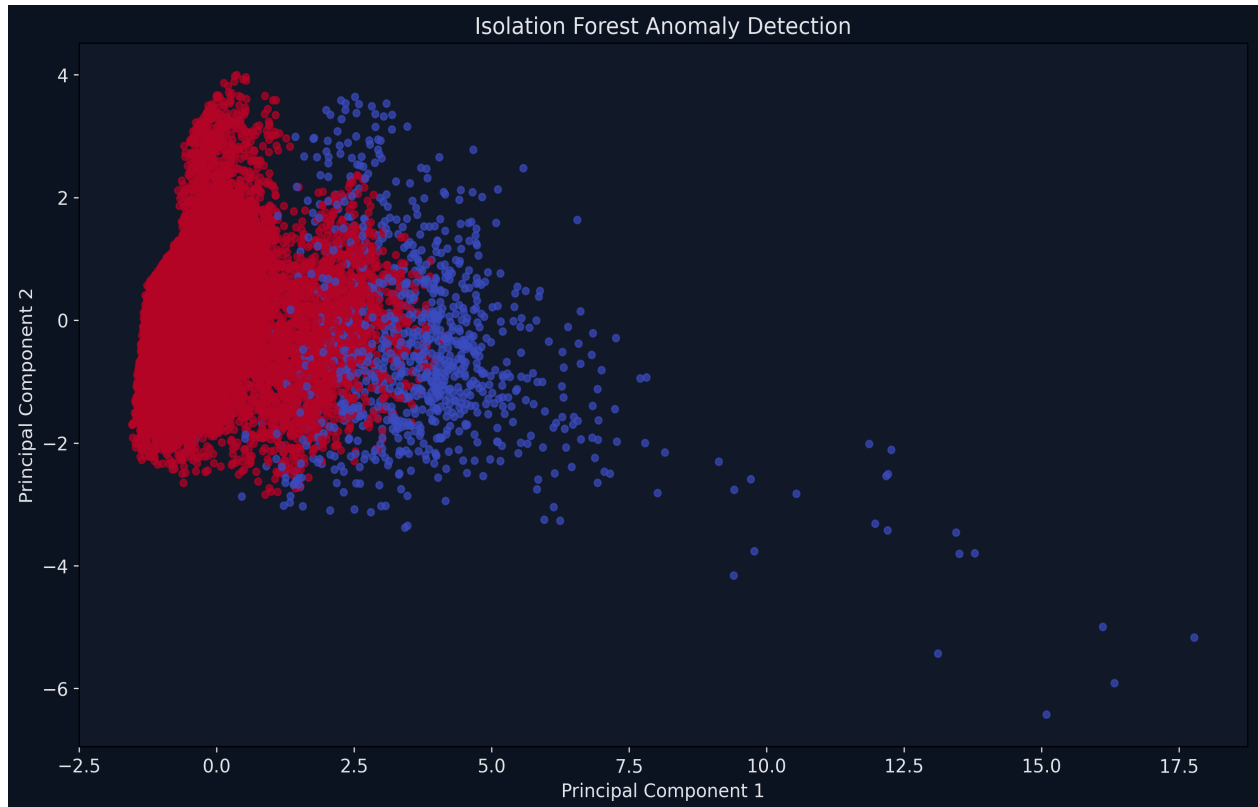
Distribution of death_flag



Axes: X-axis: death_flag | Y-axis: Frequency

Explanation: Histogram with adaptive binning revealing spread, skewness, and density variation.

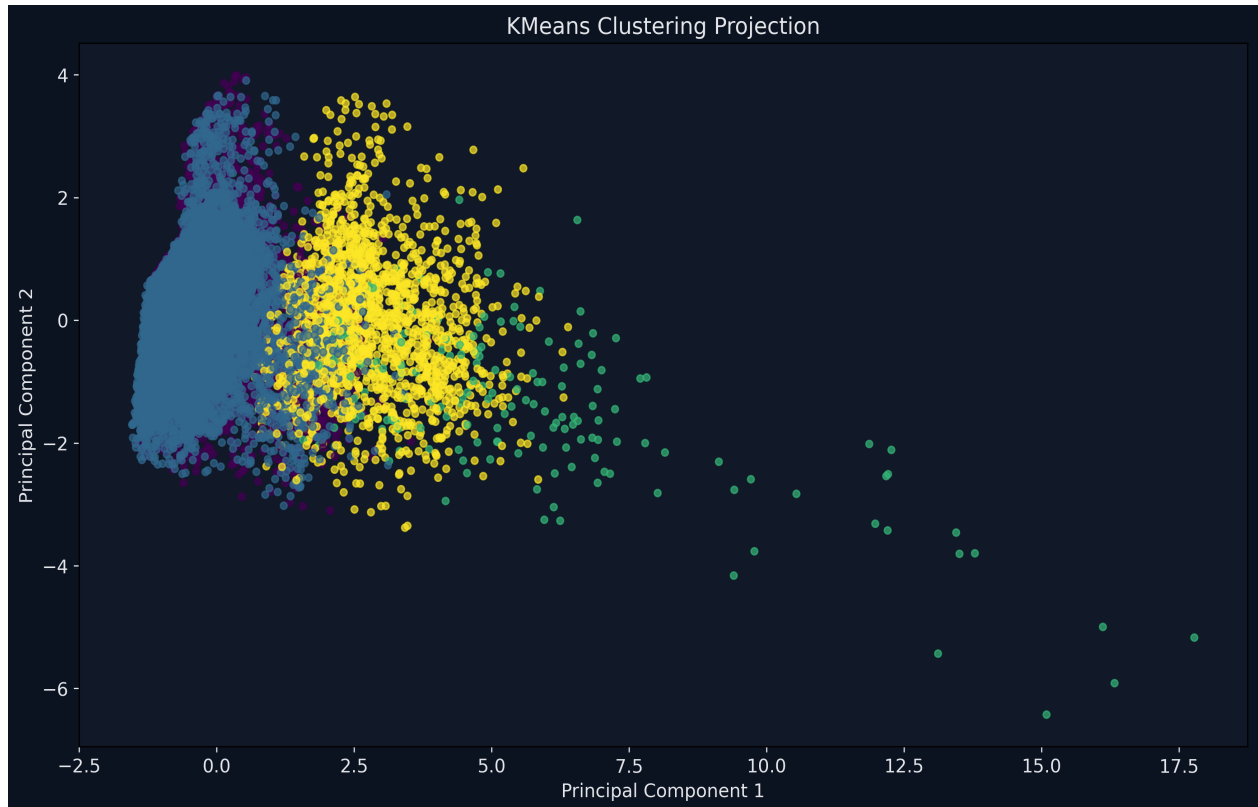
Isolation Forest Anomaly Detection



Axes: X-axis: Principal Component 1 | Y-axis: Principal Component 2

Explanation: Isolation Forest separation of anomalous observations in reduced space.

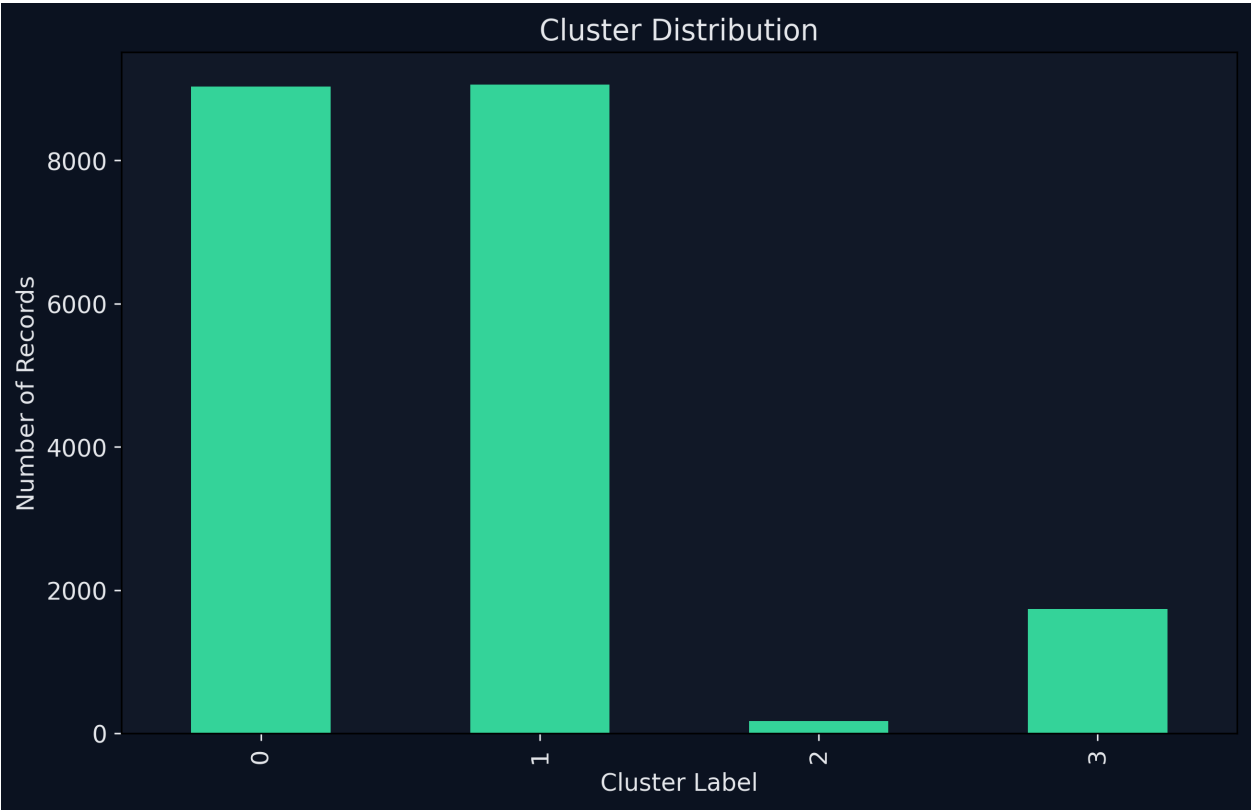
KMeans Clustering Projection



Axes: X-axis: Principal Component 1 | Y-axis: Principal Component 2

Explanation: Segmentation of data into four structural clusters.

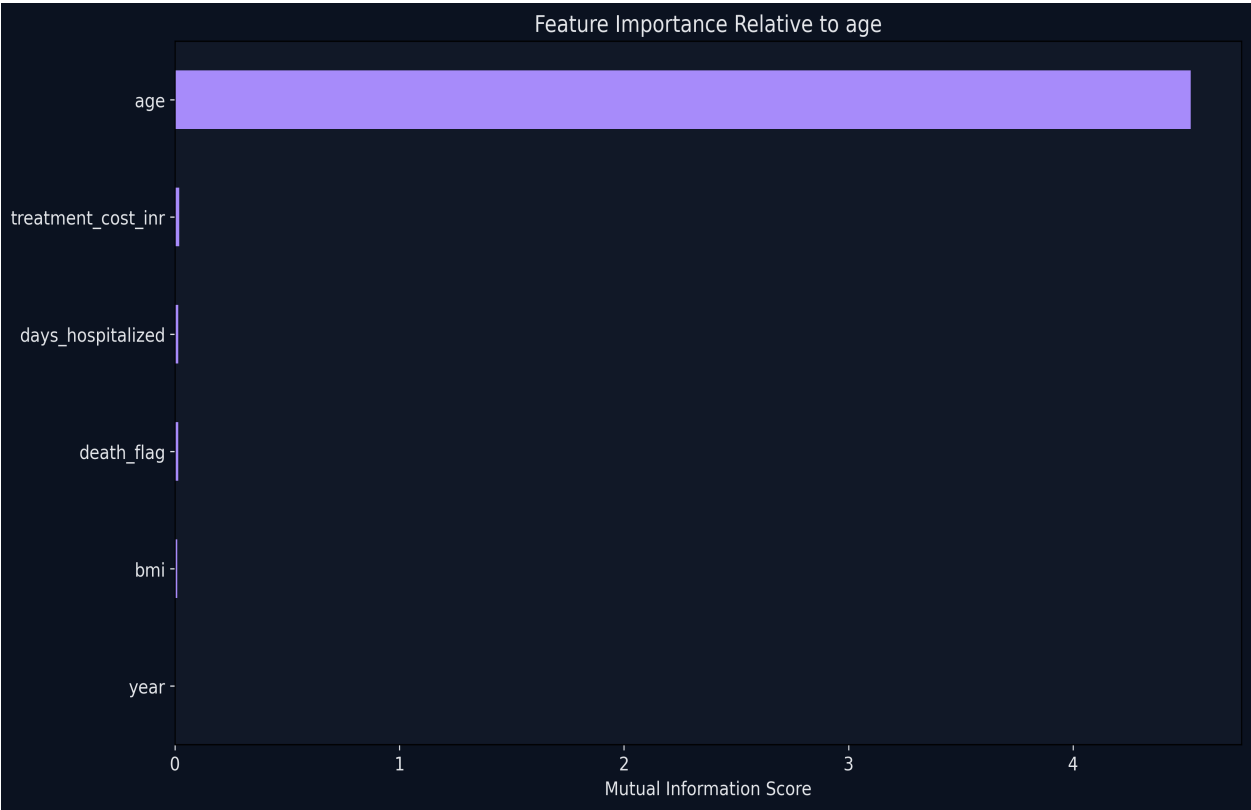
Cluster Distribution



Axes: X-axis: Cluster Label | Y-axis: Number of Records

Explanation: Population distribution across discovered clusters.

Feature Importance Relative to age



Axes: X-axis: Mutual Information Score | Y-axis: Features

Explanation: Nonlinear dependency strength between variables and reference feature.